Alexi Turcotte

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Current Position

2023–date **Postdoctoral Researcher**, *CISPA Helmholtz Center for Information Security*, Saarbrücken, Germany
Advised by Andreas Zeller. Web software supply chain security, language-based testing, and analyzers for data science languages.

Education

2018–2023 **PhD in Computer Science**, *Northeastern University*, Boston, MA, USA
Advised by Frank Tip and Jan Vitek. Thesis: *Optimizing Asynchronous JavaScript Applications*.

2016–2018 **MMath in Computer Science**, *University of Waterloo*, Waterloo, Canada
Advised by Gregor Richards. Thesis: *Reasoning About FFIs: Blame and Nondeterministic Formal Semantics*.

2012–2016 **BSc Math and Computer Science**, *Laurentian University*, Sudbury, Canada
Honours thesis advised by Hafida Boudjellaba: *Optimal Parameters for Predator-Prey Models*.

Selected Papers

ASE'26 José Antonio Zamudio Amaya, Gaetano Sapia, **Alexi Turcotte**, David Benavides, Marcel Böhme, and Andreas Zeller. *Feature-Focused Test Generation.* `https://dl.acm.org/doi/abs/10.1145/3832783.3837479`

VERIFAI'26 (Workshop) Laura Plein, **Alexi Turcotte**, Arina Hallemans, and Andreas Zeller. *Synthesizing Realistic Test Data without Breaking Privacy.*

ICSE'26 Wenyuan Xu, **Alexi Turcotte**, and Cristian-Alexandru Staicu. *D-BUNDLR: Destructing JavaScript Bundles for Effective Static Analysis.*

ICSE'26 Farideh Khalili, Satyajit Gokhale, **Alexi Turcotte**, Dale Xu, and Frank Tip. *Remediating Superfluous Re-Rendering in React Applications.* (Khalili and Gokhale are co-first authors.) **ACM SIGSOFT Distinguished Paper Award.**

ASE'25 **Alexi Turcotte** and Neev Nirav Mehta. *The Fault in our Stats.*

FSE'25 **Alexi Turcotte** and Zheyuan Wu. *Expressing and Checking Statistical Assumptions.* **ACM SIGSOFT Distinguished Paper Award.**

TSE'24 @ FSE'25 Denini Silva, Martin Gruber, Satyajit Gokhale, Ellen Arteca, **Alexi Turcotte**, Marcelo d'Amorim, Wing Lam, Stefan Winter, and Jonathan Bell. *The effects of computational resources on flaky tests.* Journal First at FSE'25.

ASE'23 **Alexi Turcotte**, Satyajit Gokhale, and Frank Tip. *Increasing the Responsiveness of Web Applications by Introducing Lazy Loading.*

ASE'22 **Alexi Turcotte**, Mark W. Aldrich, and Frank Tip. *Reformulator: Automated Refactoring of the N+1 Problem in Database-Backed Applications.*

EMSE'22 @ ICSE'23 **Alexi Turcotte**, Ellen Arteca, Ashish Mishra, Saba Alimadadi, and Frank Tip. *Stubbifier: Debloating Dynamic Server-Side JavaScript Applications.* (Turcotte and Arteca are co-first authors.) Journal First at ICSE'23.

ICSE'22 **Alexi Turcotte**, Michael D. Shah, Mark W. Aldrich, and Frank Tip. *DrAsync: Identifying and Visualizing Anti-Patterns in Asynchronous JavaScript.* **Best Artifact Award.**

OOPSLA'22 Satyajit Gokhale, **Alexi Turcotte**, and Frank Tip. *Automatic Migration from Synchronous to Asynchronous JavaScript APIs.*

OOPSLA'20 **Alexi Turcotte**, Aviral Goel, Filip Křikava, and Jan Vitek. *Designing Types for R, Empirically.*

ECOOP'19 **Alexi Turcotte**, Ellen Arteca, and Gregor Richards. *Reasoning About Foreign Function Interfaces Without Modelling the Foreign Language.*

OOPSLA'17 Gregor Richards, Ellen Arteca, and **Alexi Turcotte**. *The VM Already Knew That: Leveraging Compile-Time Knowledge to Optimize Gradual Typing.*

Other Papers

ASE'22 (Tool) Mark W. Aldrich, **Alexi Turcotte**, Matthew Blanco, and Frank Tip. *Augur: Dynamic Taint Analysis for Asynchronous JavaScript.*

SLE'22 (Tool) **Alexi Turcotte**, Pierre Donat-Bouillud, Filip Křikava, and Jan Vitek. *signatr: A Data-Driven Fuzzing Tool for R.*

MSR'22 (Tool) Ellen Arteca and **Alexi Turcotte**. *npm-filter: Automating the Mining of Dynamic Information from npm Packages.*

DMTCS'22 Ahmad Biniaz, Kshitij Jain, Anna Lubiw, Zuzana Masárová, Tillmann Miltzow, Debajyoti Mondal, Anurag Murty Naredla, Josef Tkadlec, and **Alexi Turcotte**. *Token Swapping on Trees.* (Alphabetical author order.)

VIS'20 Cameron Moy, Julia Belyakova, **Alexi Turcotte**, Sara Di Bartolomeo, and Cody Dunne. *Just typeical: Visualizing Common Function Type Signatures in R.*

ICOOOLPS'19 **Alexi Turcotte** and Jan Vitek. *Towards a Type System for R.*

CCCG'18 Therese Biedl, Ahmad Biniaz, Veronika Irvine, Philipp Kindermann, Anurag Murty Naredla, and **Alexi Turcotte**. *Integral Unit Bar-Visibility Graphs.* (Alphabetical author order.)

LATIN'18 Therese Biedl, Martin Derka, Veronika Irvine, Anna Lubiw, Debajyoti Mondal, and **Alexi Turcotte**. *Partitioning Orthogonal Histograms into Rectangular Boxes.* (Alphabetical author order.)

Awards

2026 Distinguished Reviewer Award at OOPSLA.

2026 ACM SIGSOFT Distinguished Paper Award at ICSE for *Remediating Superfluous Re-Rendering in React Applications*, awarded to the top 1.5% of submissions.

2025 ACM SIGSOFT Distinguished Paper Award at FSE for *Expressing and Checking Statistical Assumptions*.

2023 Dissertation Completion Fellowship, university-level scholarship for one or two last-term PhD students.

2022 Outstanding Service Award, Northeastern, for one PhD student with exceptional university service.

2022 Best Artifact Award at ICSE, for *DrAsync*.

2018–2021 NSERC PGS-D, highly selective national Canadian scholarship for PhD studies.

2018 Northeastern Graduate Fellowship, granted to a limited number of first-year PhD students.

CVEs and Security Advisory Credits

2026 GHSA-fq9c-6ph9-42r5: jsonpathly quadratic time on large result sets allows denial of service

2026 CVE-2026-33864: Convict has Prototype Pollution via startsWith() function

2026 CVE-2026-28491: Prototype Pollution in properties-reader

2026 GHSA-gcrg-hrj9-ggqg: Prototype Pollution via constructor.prototype in confinit

2026 CVE-2026-27212: Prototype Pollution in swiper

2026 CVE-2026-26021: set-in Affected by Prototype Pollution

2026 CVE-2026-25521: locutus is vulnerable to Prototype Pollution

2026 CVE-2026-25047: deepHas vulnerable to Prototype Pollution via constructor.prototype

2026 CVE-2026-22863: Deno node:crypto doesn't finalize cipher

Service

2027 FSE Program Committee

2026 ASE Journal First and Tools and Demos Program Committees; DEBT Program Committee; OOPSLA Program Committee; OOPSLA Web and Publicity Co-Chair

2024, 2025 ISSTA Program Committee

2023–date Reviewer for ACM TOSEM and IEEE TSE

2019–2022 Artifact Evaluation Committees: ISSTA'22, ISSTA'20, ECOOP'20, OOPSLA'19

Teaching

2026 Co-instructor for *Twelve Test Generators for Twelve Programming Languages*, a seminar I designed at CISPA

2024–date Bachelor's and Master's Thesis Seminar, CISPA

2020 Lead TA for Software Engineering, Northeastern University

2016–2018 Tutorial Instructor for CS115 (Introductory CS) and CS341 (Algorithms), University of Waterloo

2015–2016 Tutorial Instructor for COSC3106/3127 (Theory of Computing and Programming Languages), Laurentian University

Advising

To date List at `https://reallytg.github.io/students/`. Seven undergraduate theses, five master's theses, one high-school student, eight undergraduate students; mentored six PhD students.

University Service

2021–2022 Faculty Hiring Committee, Khoury College of Computer Sciences, Northeastern

2020–2021 PhD Student Admissions Committee

2018–2019 Curriculum Committee, College of Computer and Information Science, Northeastern

Invited Talks

2022–2023 *Detecting and Repairing Anti-Patterns in Asynchronous JavaScript* at University of Stuttgart, Aarhus University, CISPA, CUNY Graduate Research Day, and Simon Fraser University

Experience

2019 Research Intern, Oracle Labs, Switzerland. Dynamic software updating and hot code reloading in the Graal/Truffle framework.

2015 Software Developer, AdvanceWorx Canada. iOS apps interfacing with the FLIR One thermal camera.

Languages

Programming Rust, Java, C, C++, C#, Objective C, Swift, Python, R, MATLAB/Octave, Rocq, TypeScript and JavaScript, Racket, CodeQL

Natural English (Fluent), French (Fluent), German (Intermediate)

Misc

Github reallyTG

Citizenship Canada